

# A Simple PromiseBQP-complete Matrix Problem

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# Challenges for Quantum Computing

- find new efficient quantum algorithms  
(provide more evidence that BQP is strictly greater than BPP)
- understand the computational power and limitations of quantum computers in comparison to classical computers

# Efficient Simulation of Quantum Models

- explore what quantum models can be simulated efficiently on a classical computers; see for example:
  - *Match gates and classical simulation of quantum circuits*, Jozsa and Miyake
  - Knill-Gottesman Theorem; *Generalised Clifford groups and simulation of associated quantum circuits*, Clark, Jozsa, Linden
  - Simulating quantum circuits using the formalism of matrix product states and the formalism of contracting tensor networks; generalizations in *On the simulation of quantum circuit*, Jozsa
  - ...

# Characterizing BQP

- approximately evaluating the Jones polynomial of plat closures of braids at roots of unity, Aharonov, Jones, Landau
- approximating the Potts model at some points of the Tutte plane, Aharonov, Arad, Eban, Landau
- approximating partition functions of lattice models in certain complex-parameter regimes, Van den Nest, Dür, Raussendorf, Briegel
- approximately evaluating quadratically signed weight enumerators, Knill, Laflamme
- ...

# Language / Decision Problem

- a language  $L$  is a subset of  $\{0, 1\}^*$
- the corresponding decision problem (“language recognition”) consists in determining if
  - either  $x \in L$
  - or  $x \notin L$

for given input  $x \in \{0, 1\}^*$

# Complexity Class BQP

- BQP is the class of decision problems that can be solved by a *uniform* family of quantum circuits  $\{U_n\}$  in the following sense:

let  $x \in \{0, 1\}^n \Rightarrow$  the application of  $U_n$  to  $|x, 0\rangle$  yields

$$U_n|x, 0\rangle = \alpha_{x,0}|0\rangle \otimes |\psi_{x,0}\rangle + \alpha_{x,1}|1\rangle \otimes |\psi_{x,1}\rangle$$

such that

- $|\alpha_{x,1}|^2 \geq 2/3$  if  $x \in L$
- $|\alpha_{x,1}|^2 \leq 1/3$  if  $x \notin L$

# Promise Problem

- a promise problem  $\Pi$  is a pair  $(\Pi_{\text{YES}}, \Pi_{\text{NO}})$  of non-intersecting sets:

$$\Pi_{\text{YES}}, \Pi_{\text{NO}} \subseteq \{0, 1\}^* \text{ and } \Pi_{\text{YES}} \cap \Pi_{\text{NO}} = \emptyset$$

- the set  $\Pi_{\text{YES}} \cup \Pi_{\text{NO}}$  is called the *promise*

- standard “language recognition” problems can be cast as the special case:

$$\Pi_{\text{YES}} \cup \Pi_{\text{NO}} = \{0, 1\}^* \text{ (trivial promise)}$$



# Complexity Class PromiseBQP

- PromiseBQP is the class of promise problems  $(\Pi_{\text{YES}}, \Pi_{\text{NO}})$  that can be solved by a *uniform* family of quantum circuits  $\{U_n\}$  in the following sense:
- let  $x \in \{0, 1\}^n \Rightarrow$  the application of  $U_n$  to  $|x, 0\rangle$  yields

$$U_n|x, 0\rangle = \alpha_{x,0}|0\rangle \otimes |\psi_{x,0}\rangle + \alpha_{x,1}|1\rangle \otimes |\psi_{x,1}\rangle$$

such that

- $|\alpha_{x,1}|^2 \geq 2/3$  if  $x \in \Pi_{\text{YES}}$
- $|\alpha_{x,1}|^2 \leq 1/3$  if  $x \in \Pi_{\text{NO}}$
- nothing is required with respect to  $x \notin \Pi_{\text{YES}} \cup \Pi_{\text{NO}}$

# Complete Problems

- no complete problems are known for complexity classes such as BPP, BQP, and MA
- complete problems are only known for their promise versions

# Diagonal Entry Estimation I

- decide if
  - $(A^m)_{jj} \geq g + \epsilon b^m$
  - $(A^m)_{jj} \leq g - \epsilon b^m$
- $A$  is a sparse symmetric matrix of size  $N \times N$  with  $\|A\| \leq b$
- $j \in [1, \dots, N]$
- $m = \text{polylog}(N)$
- $\epsilon = 1/\text{polylog}(N)$
- $g \in [-b^m, b^m]$

# Diagonal Entry Estimation II

- the task is to decide if the tuple  $(A, b, m, j, \epsilon, g)$  is an element of  $\Pi_{\text{YES}}$  or of  $\Pi_{\text{NO}}$
- the instances are defined as follows:

- $(A, b, m, j, \epsilon, g)$  is a YES instance iff

$$(A^m)_{jj} \geq g + \epsilon b^m$$

- $(A, b, m, j, \epsilon, g)$  is a NO instance iff

$$(A^m)_{jj} \leq g - \epsilon b^m$$

- the promise is that only tuples in  $\Pi = \Pi_{\text{YES}} \cup \Pi_{\text{NO}}$  are considered

# DEE is PromiseBQP-complete

- Diagonal Entry Estimation is PromiseBQP-complete
- we prove this by showing that
  - DEE is in PromiseBQP  $\Leftrightarrow$   
DEE can be solved efficiently on a quantum computer
  - DEE is PromiseBQP-hard  $\Leftrightarrow$   
quantum circuits can be encoded in such a way that we can decide whether they accept an input or not by looking at the corresponding diagonal entry

# Spectral Measure I

- let

$$A = \sum_{\lambda} \lambda Q_{\lambda}$$

be the spectral decomposition of  $A$  and  $|\psi\rangle \in \mathbb{C}^N$  some unit vector

- the spectral measure induced by  $A$  and  $|\psi\rangle$  is a probability distribution on the spectrum of  $A$  such that

$$\Pr(\lambda) = \|Q_{\lambda}|\psi\rangle\|^2$$

- we refer to the process of sampling from the spectral measure as measuring the observable  $A$  in the state  $|\psi\rangle$

# Spectral Measure II

- the expected value of  $A^m$  in the state  $|\psi\rangle$  is given by

$$\langle\psi|A^m|\psi\rangle = \sum_{\lambda} \lambda^m \langle\psi|Q_{\lambda}|\psi\rangle$$

and is equal to the  $m$ th statistical moment of the spectral measure

- we have  $(A^m)_{jj} = \langle j|A^m|j\rangle$

$\Rightarrow$  find quantum algorithm for estimating statistical moments of spectral measures to show that DEE is in PromiseBQP

# Outline of Quantum Algorithm

- we measure the observable  $A$  in the state  $j$
- we obtain the eigenvalue  $\lambda$  and compute  $\lambda^m$
- we repeat this process and compute the average value of these powers



# Measuring the Observable $A$

- we can measure the observable  $A$  by
  - considering  $A$  as a Hamiltonian and simulating the corresponding time dynamics

$$U = \exp(iA/\|A\|)$$

- applying the phase estimation algorithm to  $U$

# Analysis of Error Sources

- this quantum algorithm works because the following errors can be made sufficiently small with polynomial resources only
  - errors due to imperfections (limited resolution and success probability) of phase estimation
  - statistical errors due to estimation of the expected value from the empirical mean
  - errors due to imperfect simulation of the Hamiltonian time evolution

# DEE is PromiseBQP-hard

# Equivalent Definition of PromiseBQP

- let  $U$  be a BQP quantum circuit and let  $x$  be the input
- let  $p_0$  and  $p_1$  denote the probability of finding the output qubit in state  $|0\rangle$  and  $|1\rangle$ , respectively
- elementary calculations show that

$$\langle x, 0 | U^\dagger \sigma_z^{(\text{output})} U | x, 0 \rangle = p_0 - p_1$$

- the sign of this quantity indicates if  $x \in \Pi_{\text{YES}}$  or  $x \in \Pi_{\text{NO}}$

$$p_0 - p_1 \leq -1/3 \text{ if } x \in \Pi_{\text{YES}} \text{ and } \geq 1/3 \text{ if } x \in \Pi_{\text{NO}}$$

# Forward-time Operator

- let  $V = U^\dagger \sigma_z^{(\text{output})} U$  and

$$V = V_{M-1} V_{M-2} \cdots V_0$$

the decomposition of  $V$  into elementary gates ( $M$  is odd)

- the forward-time operator  $F$  is

$$F = \sum_{\ell=0}^{M-1} |\ell+1\rangle\langle\ell|_{\text{clock}} \otimes V_\ell$$

- we have  $F^{2M} = I$  because  $V^2 = I$

# Orbit

- we restrict our attention to the orbit

$$\{F^\ell |0\rangle_{\text{clock}} \otimes |x, 0\rangle : \ell \in \mathbb{N}\}$$

# Action of the Forward-time Operator

- the input  $x$  is fully rejected ( $p_0 = 1$ )  $\Rightarrow$ 
  - the orbit is  $M$ -periodic
  - the action of  $F$  is isomorphic to the action of a cyclic shift in  $M$  dimensions

$$|\ell\rangle \mapsto |(\ell + 1) \bmod M\rangle$$

where  $F^\ell |0\rangle \otimes |x, 0\rangle$  corresponds to  $|\ell\rangle$

- the spectral measure  $R^{(0)}$  induced by  $F$  and  $|0\rangle \otimes |x, 0\rangle$  is the uniform distribution on the  $M$ th roots of unity

$$\Pr \left( \exp(i\pi 2\ell/M) \right) = \frac{1}{M}$$

# Action of the Forward-time Operator

- the input  $x$  is fully accepted ( $p_1 = 1$ )  $\Rightarrow$
- the action of  $F$  is isomorphic to the action of a cyclic shift in  $M$  dimensions with an **additional phase  $-1$**

$$\begin{aligned} |\ell\rangle &\mapsto |\ell + 1\rangle \text{ for } \ell = 0, \dots, M - 2 \\ |M - 1\rangle &\mapsto -|0\rangle \end{aligned}$$

- the spectral measure  $R^{(1)}$  induced by  $F$  and  $|0\rangle \otimes |x, 0\rangle$  is

$$\Pr \left( \exp(i\pi(2\ell + 1)/2M) \right) = \frac{1}{M}$$



# Spectral Measure

- in the general case, the spectral measure induced by  $F$  and  $|0\rangle \otimes |x, 0\rangle$  is a convex combination

$$R = p_0 R^{(0)} + p_1 R^{(1)}$$

# Sparse Matrix

- we define the self-adjoint sparse matrix

$$A = \frac{1}{2}(F + F^\dagger)$$

- the spectral measure induced by  $A$  and  $|0\rangle \otimes |x, 0\rangle$  is given by

$$P = p_0 P^{(0)} + p_1 P^{(1)}$$

where the supports of  $P^{(0)}$  and  $P^{(1)}$  are given by the real parts of  $R^{(0)}$  and  $R^{(1)}$  (cosine values)

# Spectral Measures

- $p_0 = 1 \Rightarrow$  the spectral measure  $P^{(0)}$  has the form

|                              | for $\ell = 1, \dots, M - 1$               |
|------------------------------|--------------------------------------------|
| $\lambda^{(0)} = +1$         | $\lambda_{\ell}^{(0)} = +\cos(2\pi\ell/M)$ |
| $\Pr(\lambda_0^{(0)}) = 1/M$ | $\Pr(\lambda_{\ell}^{(0)}) = 2/M$          |

- $p_1 = 1 \Rightarrow$  the spectral  $P^{(1)}$  has the form

|                              | for $\ell = 1, \dots, M - 1$               |
|------------------------------|--------------------------------------------|
| $\lambda^{(1)} = -1$         | $\lambda_{\ell}^{(1)} = -\cos(2\pi\ell/M)$ |
| $\Pr(\lambda_0^{(1)}) = 1/M$ | $\Pr(\lambda_{\ell}^{(1)}) = 2/M$          |

# Statistical Moments

- identify  $j$  with the label of the initial state  $|0\rangle \otimes |x, 0\rangle$
- the  $m$ th statistical moment is equal to

$$(A^m)_{jj} = (p_0 - p_1) \cdot \left( 1 + \sum_{\ell=1}^{M-1} \cos^m(2\pi\ell/M) \right)$$

- for  $m = M^3 = \text{poly}(\# \text{ quantum gates})$ , we have

$$(A^m)_{jj} \approx p_0 - p_1$$

# Strengthening of the Result

- the DEE problem for sparse matrices  $A$  with entries restricted to  $0, \pm 1$  remains PromiseBQP-hard
- the problem of estimating off-diagonal terms is also in PromiseBQP
- if  $A$  is restricted to have only  $0, 1$  as entries, then the problem of estimating differences

$$(A^m)_{ii} - (A^m)_{ij}$$

is PromiseBQP-complete

- the adjacency matrix  $A$  can be obtained from simple string rewriting rules (more explicit than saying that  $A$  is sparse)

# Connection to Quantum Algorithms

quantum algorithms that

- implement

$$\exp(-iAt)$$

- and compute

$$\langle \psi | f(A) | \psi \rangle$$

for sparse  $A$  and smooth functions  $f$

- *Quantum algorithm for solving linear systems of equations, Harrow, Hassidim, Lloyd*
- *A quantum algorithm to solve nonlinear differential equations, Leyton, Osborne*

# Conclusions

- a purely classical (quantum free) characterization of BQP in terms of a simple matrix problem
- quantum computers are better at counting and comparing the number of solutions of combinatorial problems (provided that BQP is strictly larger BQP)
- $\Rightarrow$  can we find new quantum algorithms based on this observation?